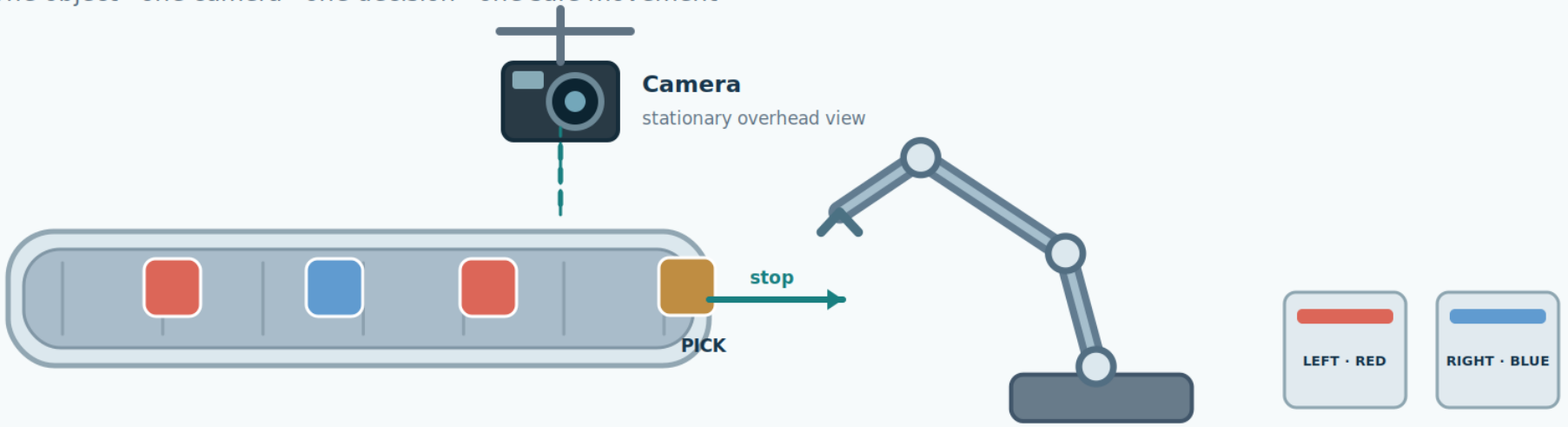
Can a robot run a factory?

A SMART SORTING FACTORY

One object · one camera · one decision · one safe movement



LEGO EV3
Motor drives the conveyor belt

PYTHON
red → left

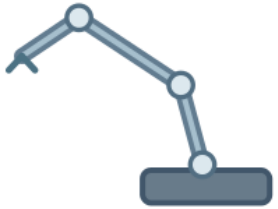
SO-ARM101 follower

SCHEMATIC ONLY · Test all positions, gripper clearance and camera field of view on real hardware.

Our miniature factory

THREE HANDS-ON STATIONS

A ROBOT



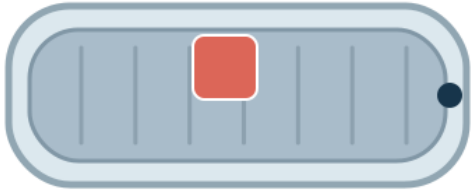
Teach and replay

B CAMERA



Recognize red / blue

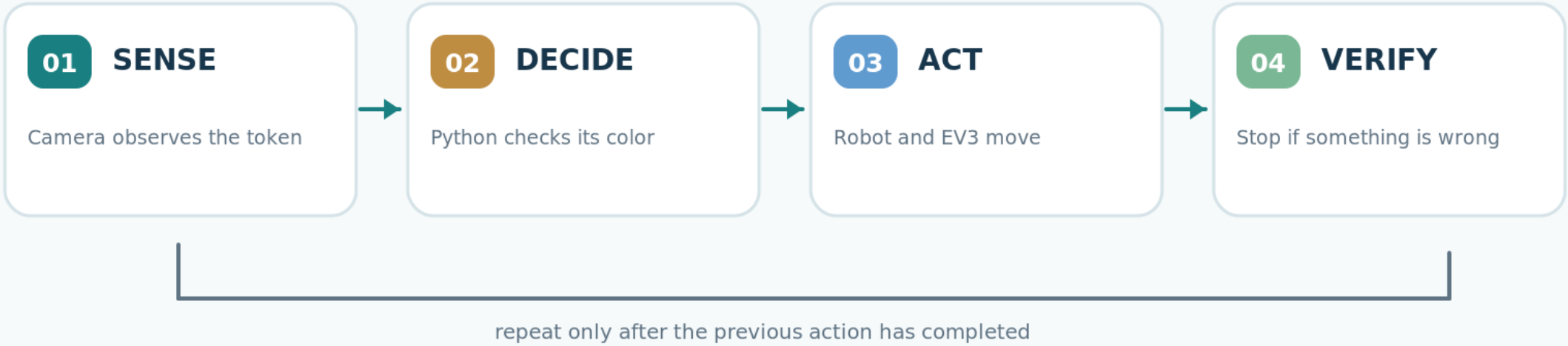
C CONVEYOR



Move and stop

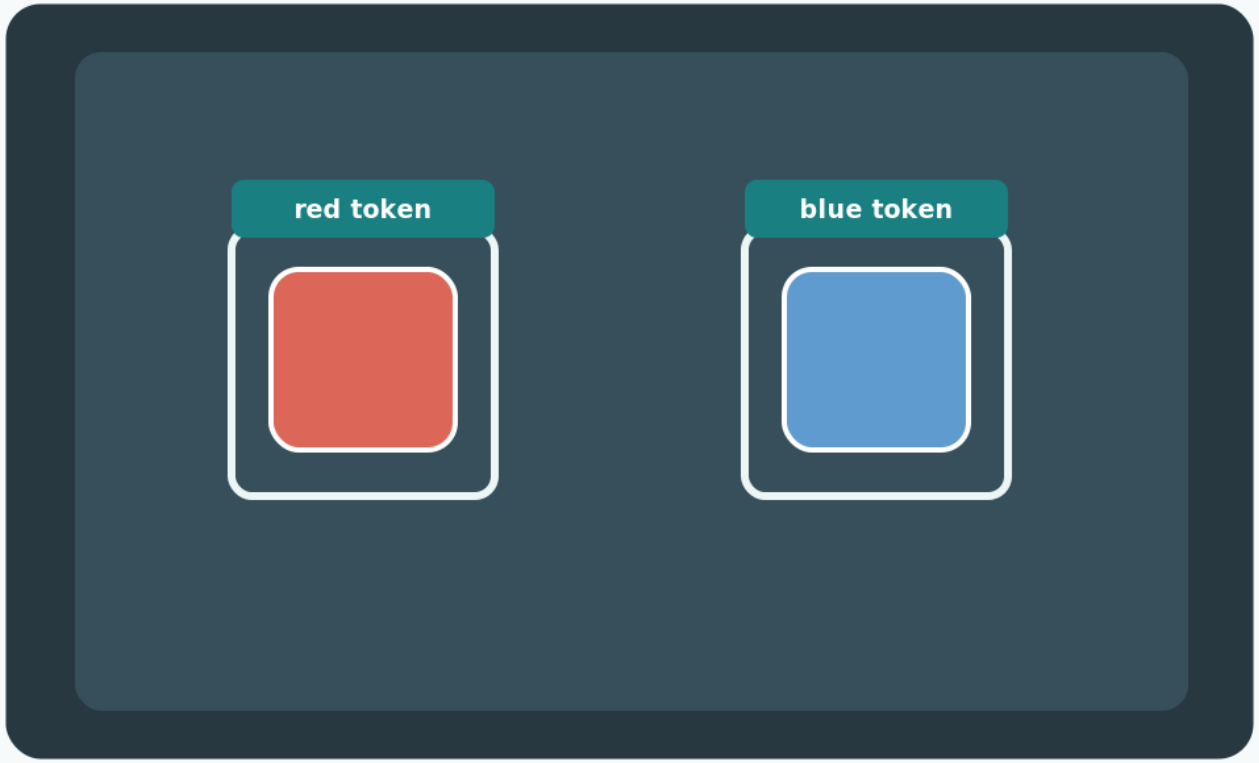
Observe, decide, act, check

THE CPS LOOP



A camera sees pixels

CAMERA PERCEPTION · SIMULATED VIEW



PYTHON RECEIVES

color = "red"

```
if color == "red":
    sort_left()
elif color == "blue":
    sort_right()
else:
    reject_unknown()
```

No GPU required for CPU color sorting.

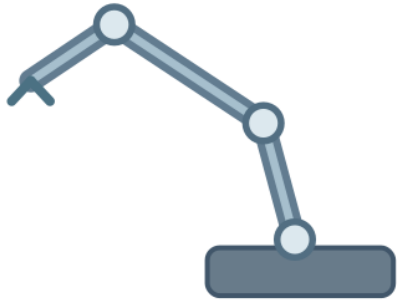
Robot joints and gripper

LEADER → FOLLOWER

Students guide a robot; the instructor supervises real motion.

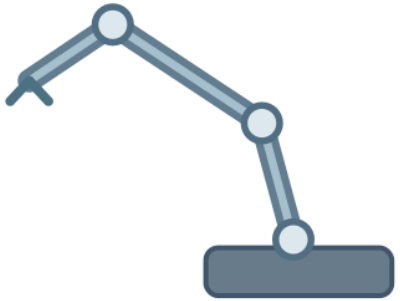
LEADER

Move a joint by hand



FOLLOWER

Repeats the taught motion



Hardware demo ONLY after calibration and safe reach test.

You can program this

```
for number in range(3):
    print("Hello robot", number + 1)

color = "red"
if color == "red":
    print("Move left")
else:
    print("Move right")
```

Try a small change

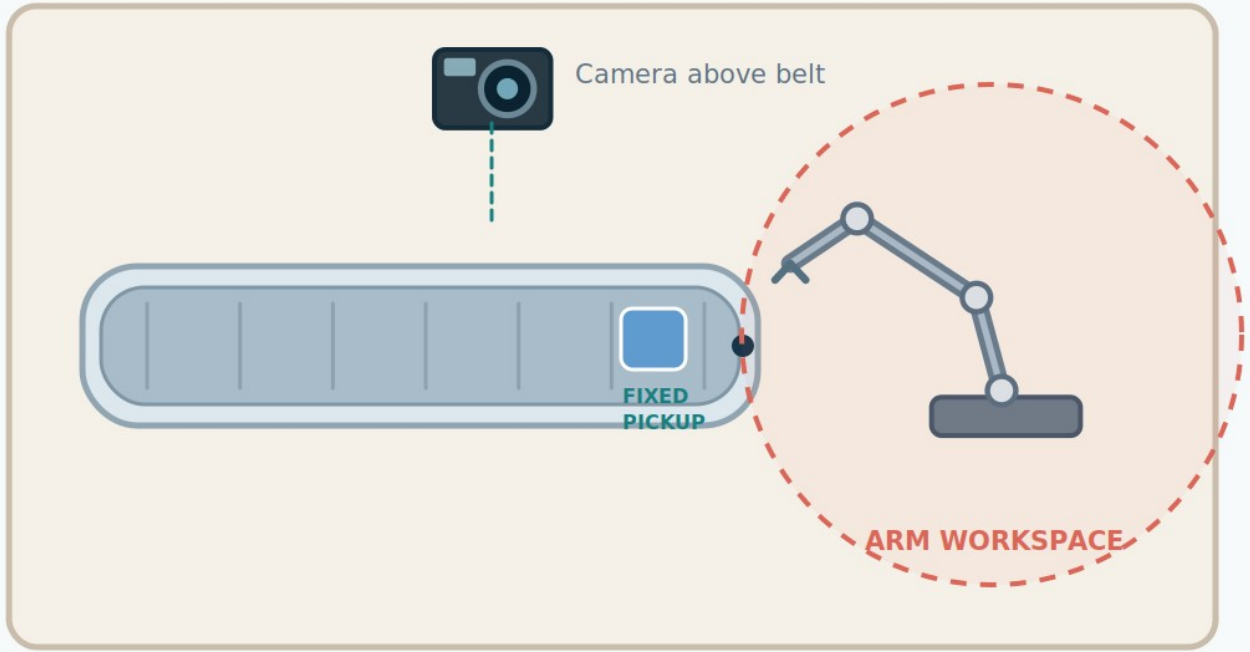
Predict the output.

Change 3 to 5.

Swap left and right.

What can go wrong?

SAFE BENCH LAYOUT · TOP VIEW



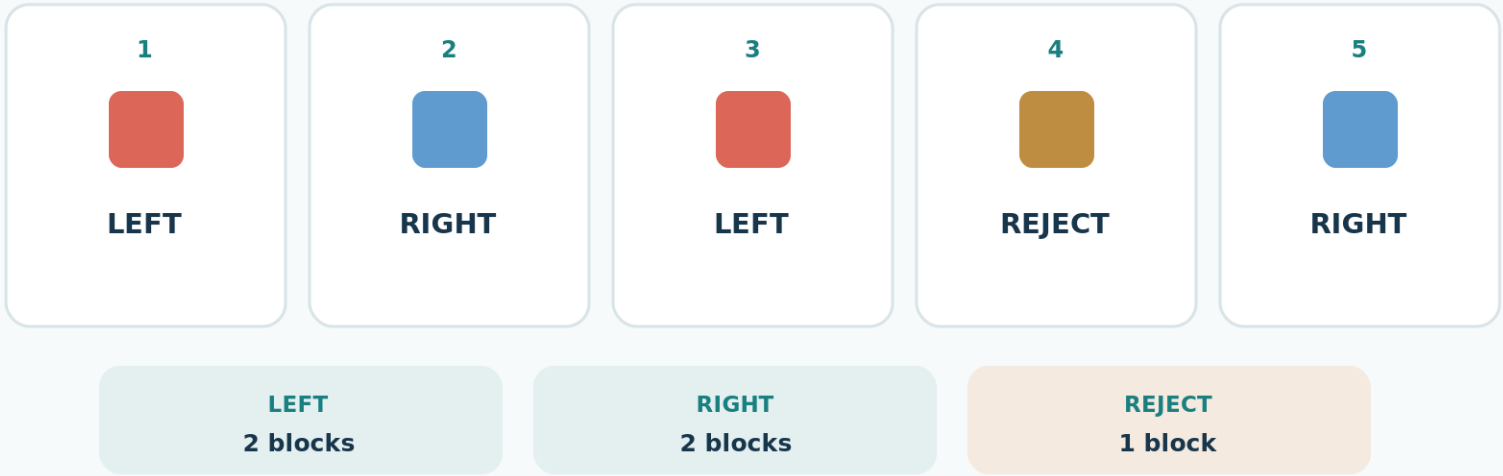
1. Stop conveyor before reaching
2. Do not enter the marked area
3. Keep a reachable power cutoff
4. Stop on missing camera frames

SCHEMATIC ONLY · Instructor must verify the actual arm sweep and emergency-stop arrangement.

Your challenge

TEST FIVE OBJECTS

Simulation input: red, blue, red, unknown, blue



LEFT	RIGHT	REJECT
2 blocks	2 blocks	1 block

Change a rule.
Test five objects.
Reject unknown.
Explain your result.

Work in pairs.
One predicts, one types.
Then swap roles.